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Quantitative Aptitude

Formula – Guide

Algebra

Algebra (Basic Identities)

Basic Identities

- $(a+b)^2 = a^2 + b^2 + 2ab$
- $(a-b)^2 = a^2 + b^2 - 2ab$
- $a^2 - b^2 = (a+b)(a-b)$
- $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$
- $(a+b)^2 - (a-b)^2 = 4ab$

Type 1 : If $x + \frac{1}{x} = n$ then

Case 1:-

- $x^2 + \frac{1}{x^2} = n^2 - 2$
- $x^4 + \frac{1}{x^4} = (n^2 - 2)^2 - 2$
- $x^8 + \frac{1}{x^8} = \left((n^2 - 2)^2 - 2\right)^2 - 2$

Case 2:-

- $x^3 + \frac{1}{x^3} = n^3 - 3n$
- $x^6 + \frac{1}{x^6} = (n^3 - 3n)^2 - 2$

Case 3:-

- $x^5 + \frac{1}{x^5} = (n^2 - 2)(n^3 - 3n) - n$
- $x^7 + \frac{1}{x^7} = (n^3 - 3n)\left[(n^2 - 2)^2 - 2\right] - n$

Type 2 : If $x - \frac{1}{x} = n$ then

- $x^2 + \frac{1}{x^2} = n^2 + 2$
- $x^3 - \frac{1}{x^3} = n^3 + 3n$
- $x^4 + \frac{1}{x^4} = \left(n^2 + 2\right)^2 - 2$
- $x^5 - \frac{1}{x^5} = \left(n^2 + 2\right)\left(n^3 + 3n\right) - n$
- $x^6 + \frac{1}{x^6} = \left(n^3 + 3n\right)^2 + 2$

Type 3 :

- $x + \frac{1}{x} = n$, Then :- $x - \frac{1}{x} = \sqrt{n^2 - 4}$
- $x - \frac{1}{x} = n$, Then :- $x + \frac{1}{x} = \sqrt{n^2 + 4}$

Type 4 : If $N = a \pm \sqrt{b}$

(where a and b are numbers):

Conditions: (i) Difference of squares of a number.

(ii) When the difference of a number is 1 . Then,

- $x + \frac{1}{x} = 2 \times$ (the bigger number)
- $x - \frac{1}{x} = 2 \times$ (the smaller number)

Some important point:

➔ $x + \frac{1}{x} = 2$ then $x = 1$

➔ $x + \frac{1}{x} = -2$ then $x = -1$

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Type 5 : Question based on $(a + b + c)^2$

- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ac)$
- $(a - b + c)^2 = a^2 + b^2 + c^2 + 2(-ab - bc + ac)$
- $(a + b - c)^2 = a^2 + b^2 + c^2 + 2(ab - bc - ac)$
- $(a - b - c)^2 = a^2 + b^2 + c^2 + 2(-ab + bc - ac)$

Type 6 : Question based on $(a + b)^3, (a - b)^3 \& (a^3 + b^3), (a^3 - b^3)$

- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
- $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Type 7 : Question based on $(a^3 + b^3 + c^3 - 3abc)$

- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)[(a + b + c)^2 - 3(ab + bc + ac)]$
- $a^3 + b^3 + c^3 - 3abc = \frac{(a+b+c)}{2}[3(a^2 + b^2 + c^2) - (a + b + c)^2]$
- $a^3 + b^3 + c^3 - 3abc = \frac{(a+b+c)}{2}[(a - b)^2 + (b - c)^2 + (c - a)^2]$
- If $a + b + c = 0$, then: $a^3 + b^3 + c^3 - 3abc = 0$
- If $a - b + c = 0$, then: $a^3 - b^3 + c^3 + 3abc = 0$
- If $a + b - c = 0$, then: $a^3 + b^3 - c^3 + 3abc = 0$
- If $a - b - c = 0$, then: $a^3 - b^3 - c^3 - 3abc = 0$

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Type 8 : Question based on Triplet ($a^2 + b^2 = c^2$)

Here are the Pythagorean triplets provided:

- $3, 4, 5$
- $5, 12, 13$
- $7, 24, 25$
- $8, 6, 10$
- $9, 40, 41$
- $8, 15, 17$
- $11, 60, 61$
- $16, 63, 65$
- $28, 45, 53$
- $39, 80, 89$
- $33, 56, 65$
- $84, 135, 159$

Type 9 : Concept Based on perfect square

We make the form of

(i) $(a + b)^2 \& (a - b)^2$

(ii) $(a \pm \alpha)^2 + (b \pm \beta)^2 + (c \pm \gamma)^2$

If you see something like this in the question then, $(a \pm \alpha) = 0, (b \pm \beta) = 0, (c \pm \gamma) = 0$

Type 10 : Question Based on value putting

Putting value = number of variable − number of equation

Polynomial

General form of a Polynomial:

$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$

Operations on Polynomial:

1. Addition: Combine like terms.
2. Subtraction: Change the signs of the terms in the polynomial being subtracted and then combine like terms.
3. Multiplication: Use distributive property or FOIL for binomials to multiply each term.
4. Division: Use polynomial long division or synthetic division to divide one polynomial by another.

Remainder Theorem:

If $P(x)$ is a polynomial and c is a constant, then:

Remainder, when $P(x)$is divided by $(x - c)$ is $P(c)$.

Factor Theorem:

If $P(x)$ is a polynomial and c is a constant, then:

- $(x - c)$ is a factor of $P(x)$ if and only if $P(c) = 0$

Quadratic Equation

Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

The sum of the roots is:

$\alpha + \beta = -\frac{b}{a}$

The product of the roots is:

$$\alpha\beta = \frac{c}{a}$$

Quadratic Equation with Roots α and β:

$$\left[x^2 - (\alpha+\beta)x + \alpha\beta\right]$$

Let P(x) = $ax^2 + bx + c$

- If $a \neq 0$, then p(x) = 0 it has 2 roots.
- If a = 0 but, $b \neq 0$ then p(x) = 0 has only 1 root.
- If a = b = c = 0 and $c \neq 0$, then p(x) = 0 has no roots, which leads to a contradiction.
- If a = b = c = 0, then p(x) = 0 has infinite roots, meaning p(x) is an identity.

Discriminant D = $b^2 - 4ac$

- **If D = 0 :** The roots are real and equal (also called double or repeated roots).
- **If D < 0 :** The roots are imaginary or complex conjugates.
- **If D > 0 :** The roots are real and distinct.

Maximum and Minimum Value of a Quadratic Equation :

- **Minimum value (when a > 0):**

$$F(x)_{\min} = \frac{4ac - b^2}{4a}$$

- **Maximum value (when a < 0):**

$$F(x)_{\max} = \frac{4ac - b^2}{4a}$$

Linear Equation

- **Linear Equation in One Variable:**

  General form: $ax + b = 0$

- **Linear equation in two variable:**

  General form: $ax + by + c = 0$

Method to solve linear equation in two variable:

1. Substitution Method: Best when one of the equations can be easily solved for one variable.
2. Elimination Method: Best when the equations can be easily manipulated to eliminate one variable.
3. Graphical Method: Useful for visualizing the solution, especially for simple systems.

System of linear equations:

General form: $a_1x + b_1y = c_1,\ a_2x + b_2y = c_2$

Types of Solutions

- **Unique Solution if:**

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

- **No Solution if:**

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \text{ and } \frac{c_1}{c_2} \neq \frac{a_1}{a_2}$$

- **Infinite Solutions if:**

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Homogeneous system of linear equations :

General form: $a_1x + b_1y = 0, \quad a_2x + b_2y = 0$

Types of Solutions

- **Only solution $x = 0, y = 0$ (trivial solution) if:**

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

- **Infinite number of solutions if:**

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

Non- Homogeneous system of linear equations :

General form: $a_1x + b_1y + c_1 = 0 \; a_2x + b_2y + c_2 = 0$

Types of Solutions

- **Parallel lines (no solution) if:**

$\frac{a_1}{a_2} = \frac{b_1}{b_2}$, but $\frac{c_1}{c_2} \neq 1$

- **Coincident lines (infinite solutions) if:**

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

- **Intersecting lines (unique solution) if:**

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$